DATA ANALYSIS REPORT

Where New York's Taxi Network Loses Money

An analysis of 39,733,818 yellow and green taxi trips taken across New York City in 2024, examining how demand, traffic speed and revenue efficiency vary by hour, zone and borough — and what changed when congestion pricing arrived in January 2025.

Prepared by **Dirgha Jivani** · Source: NYC Taxi & Limousine Commission trip records (public) · 41,829,938 raw records processed, 94.99% retained after validation · Analysis in SQL (DuckDB); figures in Python

39.7M	$1.12B	10.8	$1.96
TRIPS ANALYSED	TOTAL FARE REVENUE	AVERAGE SPEED (MPH)	REVENUE PER MINUTE
Jan–Dec 2024	$28.13 average	10.3 by December	the efficiency metric

> **The core finding:** the places and times with the most passengers are frequently the least profitable to serve. Revenue per minute across the busiest afternoon hours averages $1.82 against a $1.96 city figure, and two of the ten busiest pickup zones in New York rank among the ten worst citywide for yield. Traffic — not demand — is the binding constraint.

Summary of findings

1. **The network slowed by 11.6% over the year.** Average speed fell from 11.60 mph in January to 10.26 mph in December. The same journeys took +21.9% longer, and fare per mile rose +12.7% — passengers paid more for slower trips.
2. **Peak demand and peak earnings are 48% apart.** The most profitable hour to drive is 05:00 ($2.58 per minute at 19.4 mph). The least profitable is 15:00 ($1.74 at 9.1 mph) — which is also inside the busiest window of the day.
3. **Two of the ten busiest zones are also among the ten worst-paying.** Midtown Center takes 1.85M pickups a year at $1.80 per minute, and Penn Station manages just 8.2 mph. LaGuardia Airport, by contrast, returns $2.41 per minute — a 34% premium on the same clock.
4. **Congestion pricing produced a real but modest speed gain.** Against an untolled Manhattan control group, speeds inside the charging zone improved +2.29 percentage points and trip durations fell -3.2%. Taxi volumes inside the zone rose 14.6%, outpacing the control.

> **Why revenue per minute, and what I expected to find.** *Fare per trip rewards long journeys; trips per hour rewards short ones. Revenue per minute of occupied time is what a driver or dispatcher actually optimises. I went in expecting the interesting variation to be geographic — some boroughs simply paying better than others. It is not. Manhattan and Queens sit within 13% of each other once you control for the airports. The variation that matters is temporal, and it is much larger: 48% between the best and worst hour of the same day, in the same city, with the same fare schedule.*

NYC Taxi Operations — Demand & Efficiency

Yellow + Green · Jan–Dec 2024 · Refreshed 31 Dec 2024

Borough
- [x] Manhattan
- [x] Queens
- [x] Brooklyn
- [x] Bronx
- [x] Staten Is.

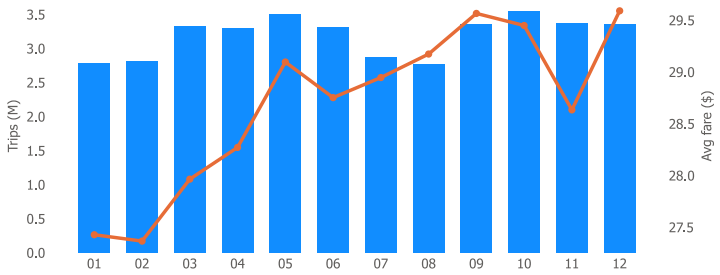
Service
- [x] Yellow
- [x] Green

Payment
- [x] Card
- [x] Cash
- [ ] Dispute
- [ ] No charge

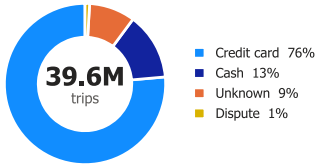
Validation
- [x] Passed only

39.7M	$1.12B	10.8	16.8	$1.96
TRIPS	REVENUE	AVG MPH	AVG MINUTES	REVENUE / MIN
94.99% validated	$28.13 avg	-11.6% Jan→Dec	+21.9% Jan→Dec	efficiency KPI

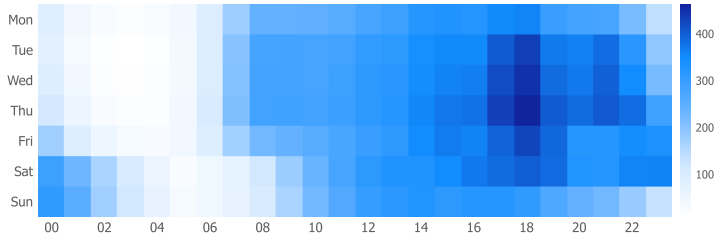
Trips and average fare by month



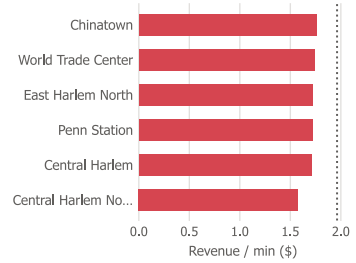
Trips by payment type



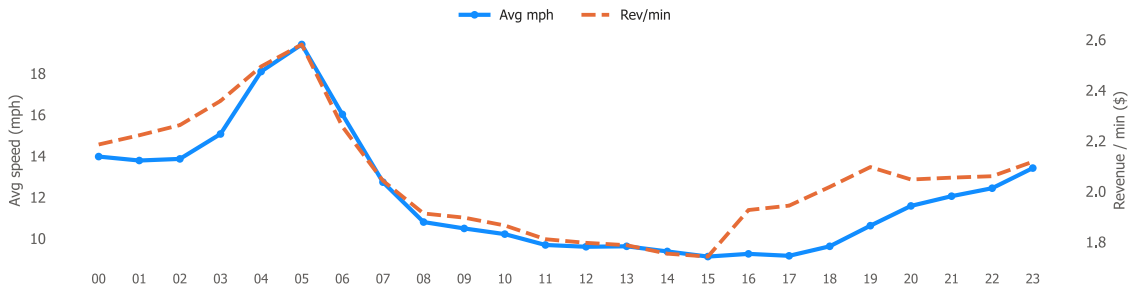
Pickup volume by weekday and hour (000s)



Lowest-yield zones



Average speed and revenue per minute by hour of day



Source: NYC TLC trip records · measures defined on page 8

> **Reading this page.** *The two charts on the bottom row are the ones that matter. Demand peaks in the early evening, but revenue per minute bottoms out in the middle of the afternoon — the busiest hours and the best-paid hours are not the same hours. Everything from page 4 onward follows from that gap.*

The city got 11.6% slower, and riders paid for it

Average taxi speed fell steadily through 2024, from 11.60 mph in January to 10.26 mph in December. This is not a distance effect — average trip length was essentially flat (27-mile-class trips throughout). What changed is how long those trips took: mean duration rose from 15.3 to 18.7 minutes, an increase of 21.9%.

Because the fare meter charges for time as well as distance, the slowdown passed straight through to price. Fare per mile rose from $7.46 to $8.40 (+12.7%), well ahead of general inflation over the same period. Passengers in December were paying meaningfully more for the same journey than passengers in January, without any change to the fare schedule.

> For an operator, this is a margin story in disguise. A driver completing a fixed number of trips per shift in January needed 22% more time to complete the same work by December — the fare rose, but so did the cost of earning it.

Monthly detail

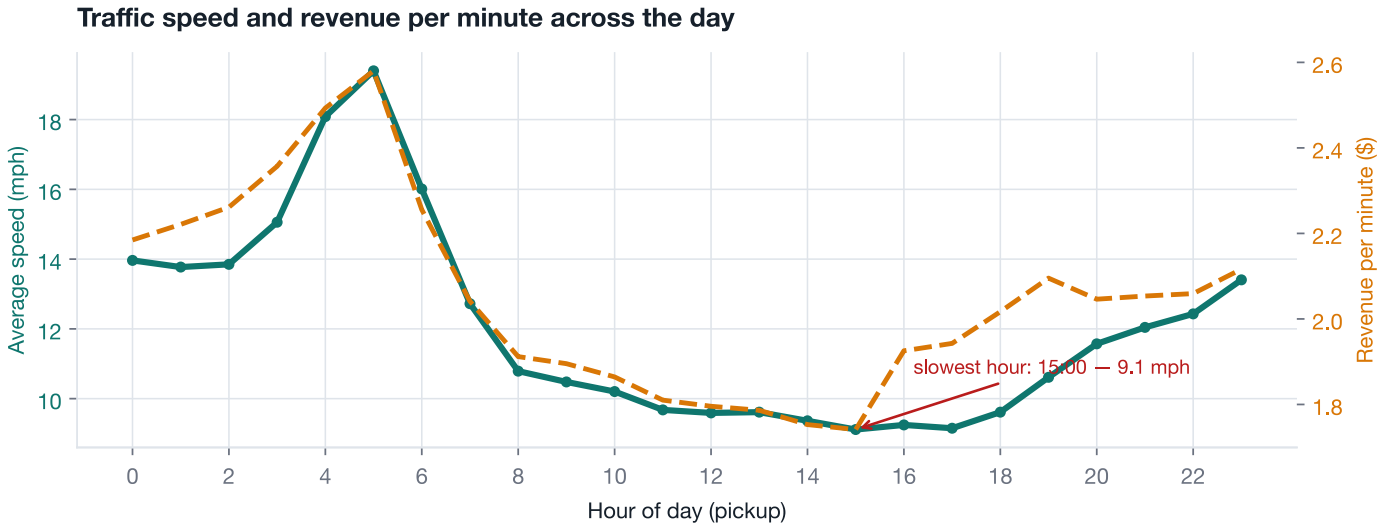
MONTH	TRIPS	AVG FARE	FARE / MILE	AVG MINUTES	AVG MPH
2024-01	2,776,509	$27.43	$7.46	15.3	11.6
2024-02	2,808,211	$27.37	$7.54	15.7	11.2
2024-03	3,326,844	$27.96	$7.56	16.5	11.3
2024-04	3,304,445	$28.27	$7.61	16.8	11.2
2024-05	3,498,804	$29.10	$7.85	17.9	10.7
2024-06	3,308,742	$28.75	$7.74	17.4	11.0
2024-07	2,868,906	$28.95	$7.56	17.0	11.5
2024-08	2,764,800	$29.17	$7.48	17.1	11.6
2024-09	3,356,776	$29.56	$7.87	18.6	10.7
2024-10	3,543,538	$29.45	$7.90	18.2	10.6
2024-11	3,375,042	$28.63	$7.95	17.7	10.6
2024-12	3,360,081	$29.59	$8.40	18.7	10.3

Trips shorter than half a mile are excluded from the fare-per-mile column, where rounding makes the ratio unstable. All other columns use the full validated dataset.

The busiest hours are the worst-paid hours

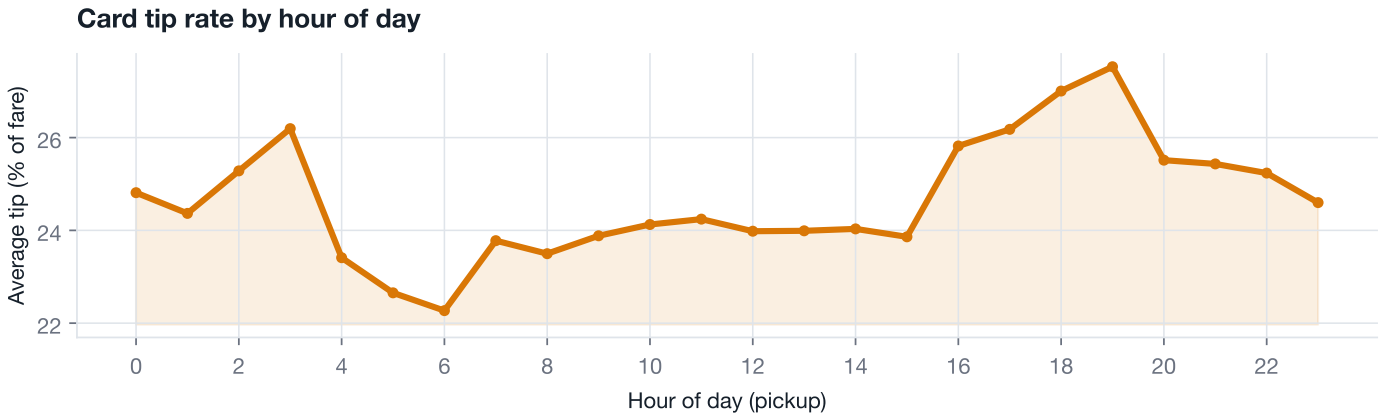
Ranking every hour of the day by revenue per minute produces an ordering almost exactly opposite to the demand curve. The best hour to be driving is 05:00, returning $2.58 per minute at 19.4 mph. The worst is 15:00 at $1.74 and 9.1 mph — a gap of 48%.

The window from 11:00 to 17:00 carries 40.8% of all trips in the city at an average of $1.82 per minute, below the $1.96 citywide figure. Traffic is the mechanism: fares in these hours are not unusually low — the average fare at 15:00 is $29.24, close to the citywide $28.13 — but it takes 19.8 minutes to earn, against 16.6 minutes at 05:00.



Revenue per minute tracks traffic speed almost perfectly. Where the city slows, earnings fall with it.

There is a supply implication. The 05:00 hour, the most profitable of the day, sees only 246,593 trips — roughly 9% of the 2,828,267 recorded at the 18:00 peak. Some of that is genuine demand thinness at dawn, and no dispatcher can conjure passengers who are asleep. But the earnings differential is large enough that the early-morning window is worth testing deliberately rather than leaving to whoever happens to be on the road.

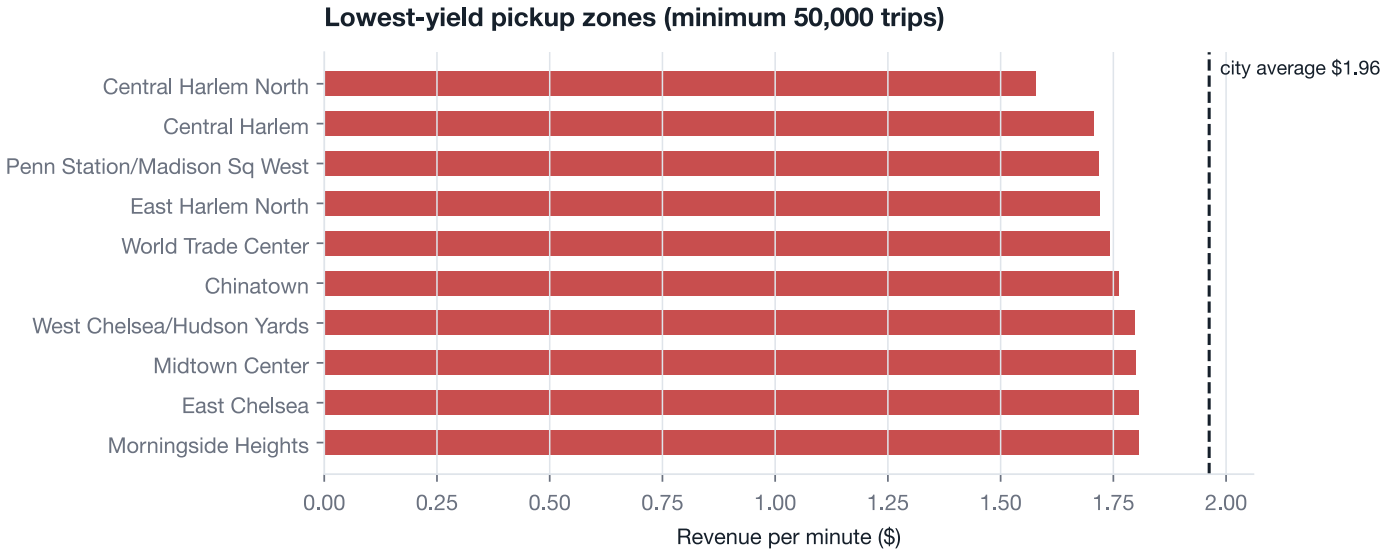


Card tips as a share of fare, by hour. The evening social peak tips best; the early-morning airport run tips worst.

Where volume and profitability part ways

Restricting to pickup zones with at least 50,000 trips a year — enough that the averages are stable — the ten lowest-yield zones in New York are all in Manhattan, and they divide into two quite different groups. Most are modest-volume zones in Upper Manhattan, where fares are small and the yield follows. The interesting cases are the other two, which are among the busiest places in the city.

Midtown Center is the second-ranked pickup zone by volume with 1,847,181 trips, yet returns only $1.80 per minute against the $1.96 city average. Penn Station/Madison Sq West records 1,311,049 pickups at 8.2 mph — the slowest high-volume zone in the dataset. These two matter operationally in a way the Harlem zones do not, because they are exactly where idle vehicles are sent.



All ten lowest-yield zones sit in Manhattan, split between Upper Manhattan and the Midtown–Chelsea core.

Lowest yield

ZONE	TRIPS	MPH	FARE	$/MIN
Central Harlem North	65,977	11.7	$23.64	**$1.58**
Central Harlem	137,786	10.9	$21.68	**$1.71**
Penn Station/Madison Sq West	1,311,049	8.2	$25.34	**$1.72**
East Harlem North	244,503	11.3	$21.35	**$1.72**
World Trade Center	206,438	11.6	$32.68	**$1.74**
Chinatown	60,925	10.7	$28.58	**$1.76**
West Chelsea/Hudson Yards	691,423	9.1	$24.68	**$1.80**
Midtown Center	1,847,181	8.5	$25.10	**$1.80**

Highest yield

ZONE	TRIPS	MPH	FARE	$/MIN
LaGuardia Airport	1,233,179	20.3	$67.31	**$2.41**
East Elmhurst	149,837	20.5	$65.19	**$2.40**
Yorkville West	739,759	11.3	$21.30	**$2.29**
Lenox Hill West	897,456	9.9	$20.96	**$2.15**
Yorkville East	540,716	11.5	$22.41	**$2.14**
Upper West Side North	764,558	10.8	$21.76	**$2.14**
Upper West Side South	1,104,796	10.4	$22.07	**$2.14**
Upper East Side North	1,684,931	9.7	$21.11	**$2.12**

The contrast is structural. High-yield zones fall into two groups: airport runs, where a long uninterrupted highway leg produces a large fare in relatively little time (LaGuardia Airport at $2.41 per minute and 20.3 mph), and Upper East Side residential zones, where trips are short but move freely. Low-yield zones are dense commercial districts where the meter runs while the vehicle does not.

Borough summary

BOROUGH (PICKUP)	TRIPS	AVG FARE	AVG MILES	AVG MPH	$/MIN
Manhattan	35,349,824	$23.78	2.41	9.7	$1.95
Queens	3,638,387	$69.65	12.11	21.3	$2.24
Brooklyn	626,629	$31.32	5.43	11.9	$1.42
Bronx	117,352	$35.22	7.66	13.8	$1.29
Staten Island	1,375	$42.42	10.11	24.9	$2.05

Queens leads on both fare and efficiency because JFK and LaGuardia sit inside it; those two zones alone account for 82% of Queens pickups. Manhattan carries 89% of all citywide volume at the lowest average speed of any borough. Staten Island appears for completeness only — at 1,375 yellow and green pickups in a full year, its averages carry no weight, and Newark is excluded from the table on the same grounds.

A real effect, smaller than the headlines

New York began charging vehicles to enter Manhattan below 60th Street on 5 January 2025. Comparing January 2025 with January 2024 isolates the change while holding the season constant, and the trip records permit a cleaner test than a simple before-and-after: because the 2025 files record the charge itself, the tolled zone can be identified from the data rather than assumed.

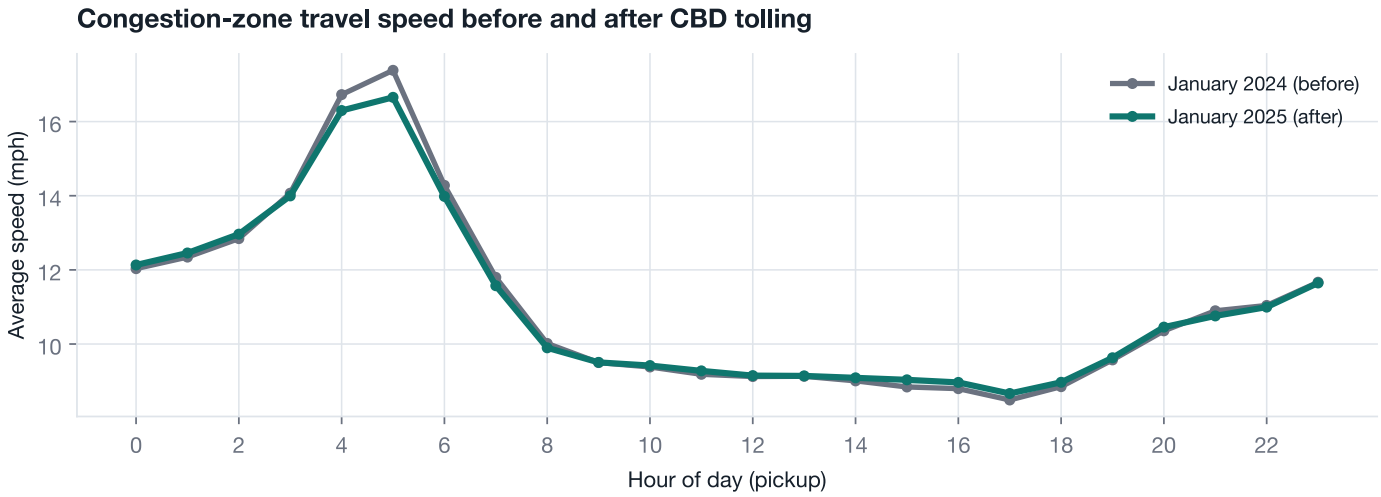
Scoring each pickup zone by the share of its trips that incurred the charge separates the city sharply — 37 zones sit near 88% and are inside the boundary, while zones north of 60th Street cluster between 16% and 30%. Those 17 northern Manhattan zones form the control group: same borough, same weather, same month, no charge.

A wrong turn worth recording

My first pass at this defined the zone as "any trip where the congestion fee was greater than zero." That looked reasonable and produced a control group of **fifteen trips**, which is what gave it away. The fee is charged on trips *entering* the zone, not on trips picked up inside it, so a fee-greater-than-zero filter quietly swept in pickups from 251 of the city's 265 zones and left almost nothing outside it to compare against. The share-based definition above is the fix. I have left the original logic in the commit history rather than tidying it away, because the failure mode — a filter that looks correct and silently defines away your control group — is the kind of thing worth being able to recognise twice.

	INSIDE CHARGING ZONE (TREATED)			NORTH OF 60TH ST (CONTROL)		
METRIC	JAN 2024	JAN 2025	CHANGE	JAN 2024	JAN 2025	CHANGE
Average speed (mph)	10.01	10.09	**+0.74%**	10.80	10.63	**-1.56%**
Average trip duration (min)	13.73	13.29	**-3.20%**	11.88	12.03	**+1.32%**
Trips completed	1,635,041	1,874,538	**+14.65%**	525,472	582,728	**+10.90%**
Average base fare ($)	15.52	15.14	-2.45%	13.90	13.91	+0.06%

Difference-in-differences on speed: +2.29 percentage points. Speeds inside the zone rose +0.74% while the untolled control fell -1.56% — so the raw before-and-after understates the effect. Trip duration inside the zone fell 3.2% while rising +1.3% in the control.



Hourly speed inside the charging zone, before and after. The change is small in absolute terms and concentrated outside the deepest congestion of mid-afternoon.

Two cautions on reading this. First, the effect is modest — a couple of percentage points against a control, not the transformation sometimes described. Second, yellow taxis are not the vehicles the policy primarily targets; they pay a small per-trip surcharge rather than the full daily charge. What this measures is the benefit taxis received from *other* traffic being priced off the road, which is arguably the cleanest available read on whether the policy moved vehicles at all.

Taxi volume inside the zone rose 14.6% year over year against 10.9% in the control, a net +3.8 points. That is consistent with some substitution from private cars toward taxis inside the boundary, though a single month cannot separate that from ordinary demand growth.

What to do with this

FLEET POSITIONING

Stop treating Midtown as the default destination for idle vehicles.

The Midtown core combines the highest pickup volume in the city with bottom-decile revenue per minute. A vehicle repositioning into Midtown Center at 15:00 is moving toward the worst earnings environment available. Upper East Side zones sit 27% higher on yield with comparable trip volume and should absorb repositioning first.

SHIFT DESIGN

Treat 04:00–06:00 as the premium window it already is.

An occupied hour worked between 04:00 and 06:00 returns $147 against $108 between 13:00 and 16:00 — a difference of $38 per hour, or 36%. That gap is not a pricing artefact; it is the same meter running against 18 mph roads instead of 9 mph ones.

Sizing it, with the caveat that matters: moving 5% of the 9,466,284 afternoon trips into the early window would be worth roughly **$5.9M a year** on the same occupied minutes. I would not present that as a target. It assumes the passengers are there to be carried, and 5% of afternoon volume is a 46% increase on everything the early window currently does — almost certainly more than latent demand at 5am can absorb. Read it as an upper bound that establishes the prize is worth a controlled test, not as a forecast.

PRICING

Recognise that the time component of the meter is already doing the work — and that it is doing it unevenly.

Fare per mile rose 12.7% across 2024 purely through congestion, with no schedule change. That protects revenue per trip but not revenue per hour, and it falls hardest on short Manhattan trips where the time component dominates. Any fare review should look at revenue per minute by zone rather than average fare, which conceals the problem.

MEASUREMENT

Adopt revenue per minute as the standard operating metric.

Average fare ranks Queens first and Manhattan last among the major boroughs; revenue per minute tells a different and more actionable story. Reporting that leads with average fare will systematically misdirect fleet decisions.

What this analysis cannot tell you

The dataset records completed metered trips only. It contains no record of time spent empty and cruising, which is where a meaningful share of driver cost sits — so revenue per *occupied* minute is an upper bound on true earnings efficiency, and zones that are easy to fill from will look worse than they are. High-volume ride-hailing trips are

reported separately and are excluded here, so this is a view of the taxi network rather than of all for-hire travel. The congestion-pricing comparison rests on a single pair of months and controls for season and borough but not for weather, fuel prices, or concurrent roadworks; it should be read as a directional indication rather than a causal estimate.

How these numbers were produced

Source. Trip-record files published monthly by the New York City Taxi & Limousine Commission, covering yellow medallion and green street-hail vehicles for January–December 2024, plus January 2025 for the congestion-pricing comparison, joined to the TLC taxi zone lookup (265 zones). Files were read directly in Apache Parquet form.

Tooling. All aggregation in SQL executed by DuckDB directly against the Parquet files; figures drawn in Python. I chose DuckDB over loading to a warehouse because 42M rows is small enough to sit on a laptop, and being able to re-run the whole waterfall in under a minute mattered more here than durability — I re-ran it four times as the validation rules changed. The pipeline is reproducible from the scripts accompanying this report: `analysis.py` writes every statistic to `results.json`, and the document interpolates from that file, so no figure here is typed by hand.

Judgement calls

Three decisions shaped the numbers more than the rest, and a reader is entitled to disagree with any of them.

The 3-hour duration ceiling. There is no natural cut-off. I settled on three hours because the distribution is effectively empty between roughly two and six hours and then spikes at exactly 24, which is a meter artefact rather than a journey. A ceiling of two hours or six changes the headline averages by less than a tenth of a mile per hour, so the choice is not load- bearing — but it should be stated rather than buried.

The 50,000-trip floor on zone rankings. Without it the lowest-yield table fills with zones recording a few hundred trips a year, where a handful of unusual journeys moves the average. The floor is arbitrary; the ranking is stable anywhere between about 20,000 and 100,000.

Tip rate as a mean of ratios. Averaging tip-over-fare across trips overweights small fares, where a routine two-dollar tip on a six-dollar ride reads as 33%. The 26% figure for Manhattan should therefore be read as a *relative* signal against other boroughs on the same measure, not as the share of revenue arriving as tips. A fare-weighted figure would be lower. I kept the simpler measure because the report uses tipping only for the shape of the hourly curve, but it would be the wrong number to put in a revenue model.

Validation waterfall

Records were removed in the sequence below, each rule applied to the output of the previous one, so losses are attributable to a specific cause. 94.99% of raw records survived.

RULE APPLIED IN SEQUENCE	REMOVED	% OF PRIOR	REMAINING
Raw records loaded (yellow + green, 2024 files)	—	—	**41,829,938**
Pickup timestamp falls inside calendar 2024	76	0.000%	**41,829,862**
Dropoff strictly after pickup	14,171	0.034%	**41,815,691**
Trip duration between 1 minute and 3 hours	536,354	1.283%	**41,279,337**
Trip distance greater than 0 and under 100 miles	456,090	1.105%	**40,823,247**
Fare and total non-negative, total greater than 0	647,574	1.586%	**40,175,673**
Total fare under $1,000 (caps data-entry outliers)	7	0.000%	**40,175,666**
Pickup and dropoff zones are known (excludes 264/265)	375,605	0.935%	**39,800,061**
Implied average speed between 1 and 70 mph	66,243	0.166%	**39,733,818**

The two largest exclusions are worth noting. Trips failing the duration bound (536,354 records) are meters left running after the passenger left, or closed within a few seconds of opening. The fare bound removes 647,574 records, of which more than 99% carry a *negative* fare_amount rather than a zero one — the signature of refunds, chargebacks and voided trips rather than of free rides. The speed bound is a final guard against distance or timestamp corruption and removes only 66,243 records (0.166%), which is a reassuring sign that the earlier filters had already caught most of the malformed rows.

Metric definitions

METRIC	DEFINITION
Revenue per minute	total_amount ÷ (trip duration in minutes). Includes tips, tolls and surcharges. The primary efficiency measure in this report.
Average speed	trip_distance ÷ (trip duration in hours). A straight-line proxy: the meter records odometer distance but not route, so this reflects realised journey speed, not free-flow road speed.
Fare per mile	fare_amount ÷ trip_distance, excluding tips and surcharges, restricted to trips of at least half a mile.
Tip rate	tip_amount ÷ fare_amount for card-paid trips only (76% of the dataset). Cash tips are not captured by the meter and are recorded as zero, so cash trips are excluded rather than counted as non-tipping.
Charging zone	Derived, not assumed: pickup zones where at least 70% of January 2025 trips incurred the CBD congestion fee. This yields 37 zones. The control group is the 17 Manhattan zones below 40%.

Representative query

Zone-level efficiency, as used for Finding 03:

```
SELECT z.zone, z.borough,
       count(*)                        AS trips,
       avg(t.mph)                      AS avg_mph,
       avg(t.rev_per_min)              AS rev_per_min,
       avg(t.total_amount)             AS avg_fare
FROM   trips t
JOIN   zones z ON t.pu_id = z.location_id
WHERE  z.borough <> 'Unknown'
GROUP  BY z.zone, z.borough
```

```
HAVING count(*) >= 50000      -- stability floor for the averages
ORDER  BY rev_per_min ASC
LIMIT  10;
```

Data accessed from the NYC TLC public trip-record archive. The Commission notes that records are submitted by technology providers and that it does not guarantee their accuracy; the validation waterfall above is the response to that caveat. Analysis and interpretation are the author’s own.